

B.s Math
6th Semester

Math 621

(Real Analysis I)

Final Term

(Preparation)

Prepared By

Amal

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Chapter 3

Continuity

Function:

A rule f that assigns to each member of a non empty set D a unique member of a set Y is a function from D to Y .

The relationship b/w a member x of D and member y of Y that f assigns to x as

$$y = f(x)$$

"The set D is the domain of f denoted by D_f ."

"The member of Y are the possible values of f ."

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range:-

if $y_0 \in Y$ and there is an x_0 in D such that $f(x_0) = y_0$ we say f attains the value y_0 .

The set of values attained by f is the range of f .

Real Valued Functions:-

The real valued function of real variable is the function whose domain and range are both subsets of the real numbers.

Example:

Range of f is $[0, \infty)$

Range of g is $[-1, 1]$

Range of h is $(0, \infty)$

The eq is

$$[f(x)]^2 = x$$

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if $x \leq 0$ no real no. is

satisfying $[f(x)]^2 = x$

while if $x > 0$ two real numbers satisfy $[f(x)]^2 = x$

So the conditions

$$[f(x)]^2 = x \text{ and } f(x) \geq 0$$

defined a function f on

$$Df = [0, \infty) \text{ with}$$

$$\text{values } f(x) = \sqrt{x}$$

Similarly the conditions

$$[g(x)]^2 = x \text{ and } g(x) < 0$$

define a function g on $Dg = [0, \infty)$

with values

$$g(x) = -\sqrt{x}$$

The ranges of f and g are $[0, \infty)$ and $(-\infty, 0]$ respectively.

Cartesian Product $X \times Y$ of two non empty sets X and Y to be the set of all ordered pairs (x, y) such that $x \in X$ and

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 $y \in Y$ thus

$$X \times Y = \{(x, y) : x \in X, y \in Y\}$$

$$\text{If } D_f \cap D_g \neq \emptyset$$

then

$$(f+g)(x) = f(x) + g(x)$$

$$(f-g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

The quotient f/g
is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Example 02

Define $(f+g)(x)$, $(f-g)(x)$,
 $(fg)(x)$ and (f/g) for the
functions $f(x) = \sqrt{4-x^2}$
and $g(x) = \sqrt{x-1}$

Solution:-

First of all we observe

$$D_f = [-2, 2] \text{ and}$$

$$D_g = [1, \infty)$$

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Then $f+g$, $f-g$ and fg are defined on $Df \cap Dg = [1, 2]$

by

$$(f+g)(x) = \sqrt{4-x^2} + \sqrt{x-1}$$

$$(f-g)(x) = \sqrt{4-x^2} - \sqrt{x-1}$$

$$(fg)(x) = (\sqrt{4-x^2})(\sqrt{x-1})$$

$$= \sqrt{(4-x^2)(x-1)}$$

The quotient f/g is defined on $(1, 2]$ by

$$\left(\frac{f}{g}\right)(x) = \sqrt{\frac{4-x^2}{x-1}}$$

f and g are not defined on $(-\infty, -2]$.

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3.1 Limits :-

The tangent line problem.
Given a function f
and a point (x_0, y_0)
on its graph, find an
equation of the line
that is tangent to the
graph at P .

The area problem.
Given a function f ,
find the area between
the graph of f and
an interval $[a, b]$ on the
 x -axis.

3.1.1 Limits (An informal view)

If the value of $f(x)$
can be made as close
as we like to L by
taking values of x
sufficiently close to a (but
not equal to a) then we
write

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$$\lim_{x \rightarrow a} f(x) = L$$

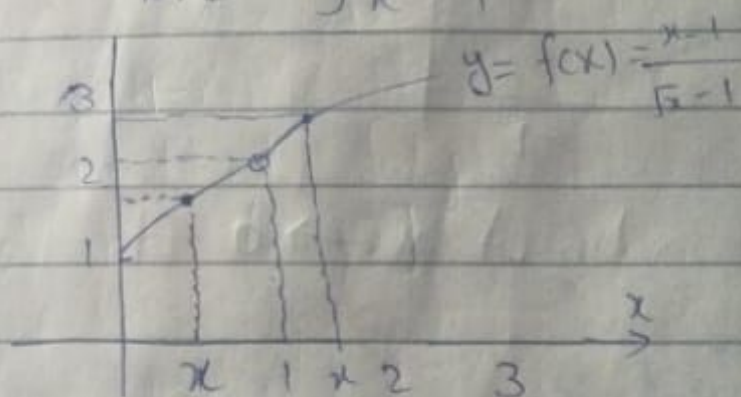
Sometimes written as

$$f(x) \rightarrow L \text{ as } x \rightarrow a$$

Example:-

investigate the limit

$$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1}$$



limit $x \rightarrow 1$ when $f(x)$ is not defined at $x=1$

Example

investigate the limit

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

limit $x \rightarrow 0$ when $f(x)$ is not defined at $x=0$

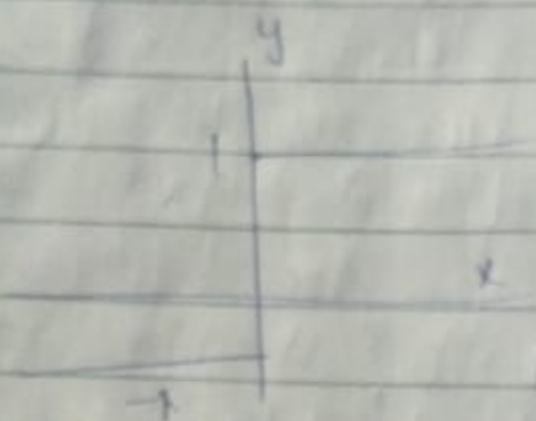
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Example

l'Hôpital's rule limit

lim $\frac{\ln x}{x}$

$$\lim_{x \rightarrow 0} \frac{\ln x}{x}$$



limit $x \rightarrow 0$

3.1.2

Formal Definition of limit

As $f(x)$ approaches the
limit L as x approaches
 x_0 and write

$$\lim_{x \rightarrow x_0} f(x) = L$$

if f is defined on
some deleted neighborhood

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of x_0 and for every $\varepsilon > 0$ there is a

$\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < |x - x_0| < \delta$$

Example

for the function defined by $f(x) = cx$ where $c \in \mathbb{R}$
show that

$$\lim_{x \rightarrow x_0} f(x) = cx_0$$

Solution

$$\begin{aligned} |f(x) - cx_0| &= |cx - cx_0| \\ &= |c| |x - x_0| \end{aligned}$$

if $c \neq 0$

$$|f(x) - cx_0| < \varepsilon$$

$$\text{if } |x - x_0| < \delta$$

where δ is any number that $0 < \delta \leq \varepsilon/|c|$

if $c = 0$ then $f(x) - cx_0 = 0$

for all x so $|f(x) - cx_0| < \varepsilon$
holds for all

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Example

Prove that $\lim_{n \rightarrow 2} (3n-5) = 1$
see lecture

$$(3(2)-5) = 1$$

$$(6-5) = 1$$

Hence proved!

Example

$$f(x) = x \sin \frac{1}{x} \quad x \neq 0$$

show that

$$\lim_{x \rightarrow 0} f(x) = 0$$

Solution:-

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f is not defined at $x_0 = 0$ because if

$$0 < |x| < \delta = \varepsilon$$

then $|f(x) - 0| = |x \sin \frac{1}{x}| \leq |x| < \varepsilon$
other hand

$$g(x) = \sin \frac{1}{x}, \quad x \neq 0$$

Most Imp Theorem

If $\lim_{x \rightarrow x_0} f(x)$ exists, then
it is unique.

Proof :-

Suppose that limit of
the function exists and
it is L_1 and L_2 .

Let $\varepsilon > 0$. from definition
of the limit there
are positive numbers δ_1
and δ_2 such that

$$|f(x) - L_i| < \varepsilon$$

$$\text{if } 0 < |x - x_0| < \delta_i$$

$$i = 1, 2$$

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if

$$\delta = \min(\delta_1, \delta_2)$$

then

$$\begin{aligned} |L_1 - L_2| &= |L_1 - f(x) + f(x) - L_2| \\ &\leq |L_1 - f(x)| + |f(x) - L_2| < 2\varepsilon \end{aligned}$$

if

$$0 < |x - x_0| < \delta$$

we have now established
an inequality that
does not depend
on x that is

$$|L_1 - L_2| < 2\varepsilon$$

Since this holds for
any positive ε

$$L_1 = L_2$$

Theorem

(for $\mathbb{R}$ or $\mathbb{C}$)

If

$$\lim_{x \rightarrow x_0} f(x) = L_1 \text{ and}$$

$$\lim_{x \rightarrow x_0} g(x) = L_2$$

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then

$$\lim_{x \rightarrow x_0} (f+g)(x) = L_1 + L_2$$

$$\lim_{x \rightarrow x_0} (f-g)(x) = L_1 - L_2$$

$$\lim_{x \rightarrow x_0} (fg)(x) = L_1 L_2$$

$$\lim_{x \rightarrow x_0} \left(\frac{f}{g} \right)(x) = \frac{L_1}{L_2} \quad L_2 \neq 0$$

Example so

$$\lim_{x \rightarrow 2} \frac{9-x^2}{x+1} \quad \text{and} \quad \lim_{x \rightarrow 2} (9-x^2)(x+1)$$

Solution

If c is any constant
then

$$\lim_{x \rightarrow x_0} c = c \quad \text{and} \quad \lim_{x \rightarrow x_0} x = x_0$$

So,

$$\lim_{x \rightarrow 2} (9-x^2) = \lim_{x \rightarrow 2} 9 - \lim_{x \rightarrow 2} x^2$$

$$= \lim_{x \rightarrow 2} (9 - (\lim_{x \rightarrow 2} x)^2) \quad (14)$$

$$= 9 - 2^2 = 5$$

and

$$\lim_{x \rightarrow 2} (x+1) = \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 1$$

$$= 2 + 1 = 3$$

Therefore

$$\lim_{x \rightarrow 2} \frac{9 - x^2}{x+1} = \frac{\lim_{x \rightarrow 2} (9 - x^2)}{\lim_{x \rightarrow 2} (x+1)}$$

$$= \frac{5}{3}$$

$$\lim_{x \rightarrow 2} (9 - x^2)(x+1)$$

$$= \lim_{x \rightarrow 2} (9 - x^2) \lim_{x \rightarrow 2} (x+1)$$

$$= 5 \cdot 3 = 15$$

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Examples -

The function

$$f(x) = 2x \sin Jx$$

Satisfy the inequality

$$|f(x)| < \varepsilon$$

whenever

$$0 < x < \delta = \varepsilon/2$$

This does not mean
that $\lim_{x \rightarrow 0} f(x) = 0$ Since
f does not define for
negative x.

3.2 One Sided Limit

$f(x)$ approaches the
left-hand limit L as
 x approaches to x_0
from the left and write

$$\lim_{x \rightarrow x_0^-} f(x) = L$$

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if f is defined on some
open interval (a, x_0)
and for each $\varepsilon > 0$
there is a $\delta > 0$
such that

$$|f(x) - L| < \varepsilon \quad \text{if}$$

$$x_0 - \delta < x < x_0$$

we say that $f(x)$
approaches the ~~to~~
right hand limit L as
 x approaches x_0 from
the right and write

$$\lim_{x \rightarrow x_0^+} f(x) = L$$

if f is defined on
some open interval
 (x_0, b) for each $\varepsilon > 0$
there is $\delta > 0$

$$|f(x) - L| < \varepsilon \quad \text{if}$$

$$x_0 < x < x_0 + \delta$$

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Theorem 3.2

A function f has a limit at x_0 if and only if it has left and right-hand limits at x_0 and they are equal.

$$\lim_{x \rightarrow x_0} f(x) = L$$

if and only if

$$\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = L$$

3.3 Limits at $\pm\infty$

If f is defined on an open interval (a, ∞) then $f(x)$ approaches the limit L as x approaches ∞

$$\lim_{x \rightarrow \infty} f(x) = L$$

For each $\varepsilon > 0$
there is number β

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Such that

$$|f(x) - L| < \varepsilon \quad \text{if } x > \beta$$

Example:-

$$f(x) = 1 - \frac{1}{x^2}$$

$$g(x) = \frac{2|x|}{1+x} \quad \text{and}$$

$$h(x) = \sin x$$

Find limits when $x \rightarrow \pm\infty$

Solution

we have

$$\lim_{x \rightarrow \infty} f(x) = 1$$

$$|f(x) - 1| = \frac{1}{x^2} < \varepsilon \quad \text{if } x > \frac{1}{\sqrt{\varepsilon}}$$

$$\lim_{x \rightarrow \infty} g(x) = 2$$

$$\text{Since } |g(x) - 2| = \left| \frac{2x}{1+x} - 2 \right|$$

$$= \frac{2}{1+x} < \frac{2}{x} < \varepsilon \quad \text{if } x > \frac{2}{\varepsilon}$$

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$\lim_{x \rightarrow \infty} h(x)$ does not exist

Since h assumes all the values between -1 and 1 in any semi-infinite interval (T, ∞)

limits $x \rightarrow -\infty$

$$f(x) = \frac{1}{x} \quad g(x) = \frac{1}{x^2}$$

$$p(x) = \sin \frac{1}{x}$$

$$q(x) = \frac{1}{x^2} \sin \frac{1}{x}$$

3.4 Infinite Limits

we say that $f(x)$ approaches ∞ as x approaches x_0 from the left and write

$$\lim_{x \rightarrow x_0^-} f(x) = \infty$$

If f is defined on an interval (a, x_0) and for each real no. M there is a $\delta > 0$ such that

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$f(x) \geq M$ if $x_0 - \delta < x < x_0$.
Similarly we can define
the following limits

$$\lim_{x \rightarrow x_0} -f(x) = -\infty \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow x_0} f(x) = \infty \quad \lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$$

$$\lim_{x \rightarrow x_0} f(x) = -\infty \quad \lim_{x \rightarrow 0^-} \frac{1}{x^2} = \infty$$

$\swarrow$
 $= \lim_{x \rightarrow 0^+} \frac{1}{x^2} = \infty$

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$

$$\lim_{x \rightarrow \infty} x^2 = \lim_{x \rightarrow -\infty} x^2 = \infty$$

Example 80

If $f(x) = e^{2x} - e^x$

we can not obtain
 $\lim_{x \rightarrow \infty} f(x)$ by writing

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} e^{2x} - \lim_{x \rightarrow \infty} e^x$$

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because this produces
indeterminate form $\infty - \infty$

So by writing

$$f(x) = e^{2x} (1 - e^{-x})$$

we find that

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \left(\lim_{x \rightarrow \infty} e^{2x} \right) \left(\lim_{x \rightarrow \infty} 1 - \lim_{x \rightarrow \infty} e^{-x} \right) \\ &= \infty (1 - 0) = \infty\end{aligned}$$

Most Imp Example

$$\lim_{n \rightarrow \infty} \frac{2n^2 - n + 1}{3n^2 + 2n - 1}$$

3.5 Continuity

Continuity (Informal)

* we say that f is continuous at x_0 if f is defined on an open interval (a, b) containing x_0 and

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

* we say that f is continuous from the left at x_0 if f is defined on an open interval (a, x_0)

$$\lim_{x \rightarrow x_0^-} f(x) = f(x_0)$$

* we say that f is continuous from the right at x_0 if f is defined on an open interval (x_0, b)

$$\lim_{x \rightarrow x_0^+} f(x) = f(x_0)$$

(23) Continuity (Formal)

- * A function f is continuous at x_0 if and only if f is defined on an open interval (a, b) containing x_0 and for each $\varepsilon > 0$ there is a $\delta > 0$ such that
- $$|f(x) - f(x_0)| < \varepsilon$$
- whenever

$$|x - x_0| < \delta.$$

- * A function f is continuous from right at x_0 if and only if f is defined on an interval $[x_0, b)$ for each $\varepsilon > 0$ there is a $\delta > 0$ such that
- $$|f(x) - f(x_0)| < \varepsilon$$
- whenever

$$x_0 \leq x < x_0 + \delta$$

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* A function f is continuous from the left at x_0 if and only if f is defined on $\cap$ interval $(a, x_0]$ and for each $\varepsilon > 0$ there is $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon \text{ whenever } x_0 - \delta < x \leq x_0$$

Example ~~23~~

Discuss the continuity of f defined on $[0, 2]$ by

$$f(x) = \begin{cases} x^2, & 0 \leq x < 1 \\ x+1, & 1 \leq x \leq 2 \end{cases}$$

Solution

$$\lim_{x \rightarrow 0^+} f(x) = 0 = f(0)$$

$$\lim_{x \rightarrow 1^-} f(x) = 1 \neq f(1) = 2$$

$$\lim_{x \rightarrow 1^+} f(x) = 2 = f(1)$$

$$\lim_{x \rightarrow 2^-} f(x) = 3 = f(2)$$

Therefore f is continuous from the right at 0 and 1 and continuous from the left at 2 but not at 1.

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Example ~~gs~~ V. Imp ϵx

Discuss the continuity

$$f(x) = \sqrt{x}$$

$$0 < x < \infty$$

Solution

$$|f(x) - f(0)| = \sqrt{x} < \epsilon$$

$$\text{if } 0 < x < \epsilon^2$$

$$\text{So } \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\text{if } x_0 > 0 \text{ and } x \geq 0$$

then

$$|f(x) - f(x_0)| = |\sqrt{x} - \sqrt{x_0}|$$

$$= \frac{|x - x_0|}{\sqrt{x} + \sqrt{x_0}} < \frac{|x - x_0|}{\sqrt{x_0}} < \epsilon$$

$$\text{if } |x - x_0| < \epsilon \sqrt{x_0}$$

$$\text{So } \lim_{x \rightarrow x_0} f(x) = f(x_0)$$

So f is continuous on $[0, \infty)$

Example ~~gs~~ MCQS

Discuss the continuity

$$g(x) = \frac{1}{\sin \pi x}$$

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Solution

The function is
continuous on
 $S = \bigcup_{n=-\infty}^{\infty} (n, n+1)$

function g is discontinuous on
any $x_0 = n$ (integer) since
it is not defined at
such points.

3.6 Piecewise Continuous Functions

A function is piecewise
continuous on $[a, b]$ if

* $\lim_{x \rightarrow x_0^+} f(x)$ exists for all
 x_0 in $[a, b]$

* $\lim_{x \rightarrow x_0^-} f(x)$ exists for all x_0
in $(a, b]$

* $\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = f(x_0)$
for all but finitely many
points x_0 in (a, b) .

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Theorem
Proof

$$\lim_{x \rightarrow x_0} (f+g)(x) = (f+g)(x_0) \\ = f(x_0) + g(x_0)$$

$$\lim_{x \rightarrow x_0} (fg)(x) = (fg)(x_0) \\ = f(x_0) g(x_0)$$

$$\lim_{x \rightarrow x_0} f(x) = L_1 \quad \rightarrow f(x_0)$$

$$\lim_{x \rightarrow x_0} g(x) = L_2 \quad \rightarrow g(x_0)$$

Example

Discuss the continuity of the following function

$$y(x) = \frac{9-x^2}{x+1}$$

$$y(x) = \frac{f(x)}{g(x)}$$

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$$f(x) = f_1(x) - f_2(x) = 9 - x^2$$

$$f(x) = 9 - x^2$$

$$f(x) = f_1(x) - f_2(x)$$

Let

$$g(x) = x + 1$$

$$= g_1(x) + g_2(x)$$

$$= x + 1$$

as $g(x) \neq 0$

it is not continuous on
 $x = -1$

$r(x)$ is continuous to f/g

3.7 Removable Discontinuity

It can be shown that

if $f_1, f_2, \dots, f_n$ are
continuous on set S then

So are $f_1 + f_2 + \dots + f_n$

and $f_1 f_2 \dots f_n$

So any rational function

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$$r(x) = \frac{a_0 + a_1x + \dots + a_nx^n}{b_0 + b_1x + \dots + b_mx^m}$$

is continuous for all values of x except those for which its

V. Imp Example

If $f(x) = \log x$

$$g(x) = \frac{1}{1-x^2}$$

Can we define $f \circ g$?

Solution:

The domain of the given functions are

$$D_f = (0, \infty) \text{ and}$$

$$D_g = x \in \mathbb{R} / x \neq \pm 1$$

Since $g(x) > 0$ if $x \in T = (-1, 1)$

The composite function $f \circ g$ is defined on $(-1, 1)$

by

$$(f \circ g)(x) = \log \frac{1}{1-x^2}$$

Can we define $g \circ f$?

The function $g \circ f$ is
define on $(0, 1/e) \cup (1/e, e) \cup (e, \infty)$
by

$$(g \circ f)(x) = \frac{1}{1-(\log x)^2}$$

Composition of functions:-
Define ~~as~~

Theorem

Suppose that g is continuous
at x_0 , $g(x_0)$ is an
interior point of Df and
 f is continuous at $g(x_0)$.
Then $f \circ g$ is continuous at
 x_0 .

Proof

Since $g(x_0)$ is an interior
point of Df and f
is continuous at $g(x_0)$

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Check the Continuity
of $f \circ g$

Solution:

The function f is continuous
for $x > 0$ the function
 g is continuous
for $x \neq -1$

Since $g(x) > 0$ if $x < -3$ or
 $-1 < x < 3$ the composition
function

$$(f \circ g)(x) = \frac{\sqrt{a-x^2}}{\sqrt{x+1}}$$

is continuous on $(-\infty, -3) \cup (-1, 3)$

It is also continuous
from the left at -3 and 3

3.8 Bounded Functions

A function is bounded
below on set S if there
is a real number m
such that

$$f(x) \geq m \text{ for all } x \in S$$

The Set

$$V = \{ f(x) : x \in S \}$$

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has an infimum a
and write

$$a = \inf_{x \in S} f(x)$$

or

$$a = \min_{x \in S} f(x)$$

Bounded above function

A function is bounded above on S if there is a real number M such that $f(x) \leq M$ for all x in S .

If V has supremum β we write

$$\beta = \sup_{x \in S} f(x)$$

or

$$\beta = \max_{x \in S} f(x)$$

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Most Imp Theorem

If f is continuous on a finite closed interval $[a, b]$, then f is bounded on $[a, b]$.

3.9 The Intermediate Value Theorem

Statement + Proof

3.10 Uniform Continuity

A function f is continuous on subset S of its domain if for each $\epsilon > 0$ and each x_0 in S , there is a $\delta > 0$ which may depend upon x_0 as well as ϵ such that

$$|f(x) - f(x_0)| < \epsilon$$

if $|x - x_0| < \delta$ and $x \in D_f$

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A function uniformly continuous on a subset S of its domain if for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(x')| < \epsilon$$

whenever

$$|x - x'| < \delta$$

and $x, x' \in S$

Example

check uniform continuity

$$f(x) = 2x$$

Solution

$$\text{for } f(x) = 2x$$

we have

$$|f(x) - f(x')| = 2|x - x'| < \epsilon$$

$$\text{if } |x - x'| < \epsilon/2$$

function $f(x)$ is uniformly continuous on $(-\infty, \infty)$

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Example

Show that the function $g(x) = x^2$ is uniformly continuous on $[-r, r]$ for any finite r .

Solution

$$\begin{aligned} |g(x) - g(x')| &= |x^2 - (x')^2| \\ &= |x - x'| \leq 2r |x - x'| \end{aligned}$$

So

$$|g(x) - g(x')| < \varepsilon \quad \text{if } |x - x'| < \delta = \frac{\varepsilon}{2r} \quad \text{and } -r \leq x, x' \leq r$$

Example

The function $g(x) = x^2$ is not uniformly continuous on $(-\infty, \infty)$.

Solution

if $\delta > 0$ there are real numbers x and x' such that

$$|x - x'| = \delta/2 \quad \text{and}$$

$$|g(x) - g(x')| = |x - x'| |x + x'|$$

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$$\text{if } |x - x'| = \delta/2$$

$$\text{and } x, x' > 1/8$$

then

$$|x - x'| |x + x'| > \frac{\delta}{2} \left(\frac{1}{8} + \frac{1}{8} \right) = \delta$$

Example:

The function

$$f(x) = \cos \frac{1}{x}$$

is continuous on $(0, 1]$ However f is not uniformly continuous on $(0, 1]$

$$\left| f\left(\frac{1}{n\pi}\right) - f\left(\frac{1}{(n+1)\pi}\right) \right| = 2$$

$$n = 1, 2, \dots$$

3.11 Monotonic Functions

A function is said to be non-decreasing on an interval I if

$$f(x_1) \leq f(x_2) \text{ whenever } x_1, x_2 \in I, x_1 < x_2 \quad (3-12)$$

or nonincreasing on I if

$$f(x_1) \geq f(x_2) \text{ whenever } x_1 \text{ and } x_2 \text{ are in } I \text{ and } x_1 < x_2 \quad (3-13)$$

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In either case, f is on I

If $\leq$ can be replaced by

$<$ in (3.12) f is

increasing on I . If $\geq$

can be replaced by

$>$ in (3.13) f is

decreasing on I .

In either of these

two cases f is strictly

monotonic on I .

Date: _____

Chapter # 04

Differentiability

Derivative:-

A function f is differentiable at an interior point x_0 of its domain if the difference quotient

$$\frac{f(x) - f(x_0)}{x - x_0}, \quad x \neq x_0,$$

if it approaches a limit as x approaches x_0 , in which case the limit is called the derivative of f at x_0 , and is denoted by $f'(x_0)$.

$$\text{So, } f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

If we take $x = x_0 + h$ then

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

! See headlines from handouts.

Example :-

If n is a positive integer, find the derivative of function $f(x) = x^n$

Sol

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x^n - x_0^n}{x - x_0}$$

$$= \frac{x - x_0}{x - x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k;$$

$$f'(x_0) = \lim_{x \rightarrow x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k.$$

$$= n x_0^{n-1}$$

Since this hold x_0 ,

$$f'(x) = n x^{n-1}$$

$$\underline{\underline{\frac{d}{dx}(x^n) = n x^{n-1}}}$$

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Example:

Find derivative of line $y = mx + c$

See page no: 112

One sided derivative:-

R-H-D If f is defined on $[x_0, b)$, then right hand derivative of f at x_0 is $f'_+(x_0) = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}$

L-H-D : If f is on $(a, x_0]$ the left hand derivative of f at x_0 is, $f'_-(x_0) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}$

→ f is differentiable on open interval (a, b) , $f'_+(a)$ and $f'_-(b)$ both exist.

→ f is continuously differentiable on $[a, b]$, f' is continuous on (a, b)

$$f'_+(a) = \lim_{x \rightarrow a^+} f'(x)$$

$$f'_-(b) = \lim_{x \rightarrow b^-} f'(x)$$

Extreme values of f

→ f is local extreme value of f if $\delta > 0$ such that $f(x) - f(x_0)$ does not change sign -- $(x_0 - \delta, x_0 + \delta) \cap D_f$

→ $f(x_0)$ is local maximum if $f(x) \leq f(x_0)$

→ $f(x_0)$ is local minimum if $f(x) \geq f(x_0)$

According to Rolle's Theorem

→ if $\neq$ two extreme values are same then f is constant on (a, b) , $f'(x) = 0$.

→ If extreme value are differ then one must be attained at some point in open interval (a, b) , $f'(c) = 0$

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Indeterminate formspg: 115

Example:

Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

Sol: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{d/dx (\sin x)}{d/dx (x)}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{\cos x}{1} = \frac{1}{1} = 1$$

Imp Example: $\lim_{x \rightarrow \infty} \frac{e^{-x}}{x}$

By applying L. Hospital's rule ---
and we know ---

$$e^{-(-\infty)} = e^{\infty} \text{ if } e^{\infty} \text{ then } e^{\infty} = \frac{\infty}{\infty}$$

So,

$$\lim_{x \rightarrow -\infty} = \frac{-e^{-x}}{1} = -\infty$$

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M T W T F S

Example:

$$\lim_{x \rightarrow +\infty} \frac{x^{-4/3}}{\sin(4/x)}$$

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Indeterminate forms:-

A product of function $f \cdot g$ is said to have the indeterminate form $0 \cdot \infty$ as $x \rightarrow b \rightarrow +\infty$ as $x \rightarrow b \rightarrow -\infty$.

Example of (117) $\lim_{x \rightarrow 0^+} x \log x$

→ Infinity - Infinity form $\infty - \infty$

A difference $f - g$ is of the form $\infty - \infty$ as $x \rightarrow b$ if

$$\lim_{x \rightarrow b-} f(x) = \lim_{x \rightarrow b-} g(x) = \pm \infty$$

Example:-

$$\lim_{x \rightarrow 0} \frac{\sin x}{x^2} = \frac{1}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x^2} = \frac{1}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{1}{x} = \lim_{x \rightarrow 0} \frac{1}{x}$$

$$= 1 \times \frac{1}{0} = \frac{1}{0}$$

So,

$$\infty - \infty$$

So, we see it is infinity form

Sol

Now,

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^2} = \frac{0}{0} \text{ form}$$

We can apply L.Hospital rule,

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M T W T F S

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{2x} = \frac{1-1}{0} = \frac{0}{0}$$

Again apply rule :-

$$\lim_{x \rightarrow 0} \frac{-\sin x + 0}{2} = \frac{0}{2} = 0$$

Imp

$$\lim_{x \rightarrow \infty} x^2 - x$$

- polynomial $x^2 - x$

$$\lim_{x \rightarrow \infty} x \left(1 - \frac{1}{x}\right)$$

$$= \infty \left(1 - \frac{1}{\infty}\right) \Rightarrow \infty (1 - 0)$$

$$= \infty$$

→ Indeterminate forms: $0, 1^\infty, \infty^0$ The function f^g is defined by

$$f(x)^{g(x)} = e^{g(x) \log f(x)} = \exp^{\text{exponent}}(g(x) \log f(x))$$

for all x such that $f(x) > 0$. (pg 118)Imp Example:-

$$\lim_{x \rightarrow 0^+} x^x$$

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{x \log x}$$

$$= e^{0 \log x}$$

$$= e^0 \Rightarrow \text{and } e^0 = 1$$

$$\therefore \log x = \infty$$

monotonic on I .

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Chapter 5

Riemann Integration

5.1

Riemann Sums

Partition of an interval:

Let f be a real valued function defined on a finite interval $[a, b]$

A partition of $[a, b]$ is a set of subintervals

$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$

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where

$$a = x_0 < x_1 < \dots < x_n = b$$

Thus any set of $n+1$ points satisfying $a = x_0 < x_1 < \dots < x_n = b$ defines a partition P of $[a, b]$ which we denote by

$$P = [x_0, x_1, \dots, x_n]$$

The points $x_0, x_1, \dots, x_n$ are the partition points of P .

The largest of lengths of the subintervals $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ is the norm of P written as $\|P\|$; thus

$$\|P\| = \max_{1 \leq i \leq n} (x_i - x_{i-1})$$

Definition of Riemann Sum

5.2 Riemann Integral

Theorem (Most Imp)

Prove that the Riemann integral $\int_a^b f(x) dx$ if it exists, is unique.

(Imp) (48)

Example

Determine if f is Riemann integrable over the given interval or not? Find $\int_a^b f(x) dx$

where $f(x) = 1$
 $a \leq x \leq b$

Solution is

For the given function

$$\sum_{j=1}^n f(c_j)(x_j - x_{j-1}) = \sum_{j=1}^n (x_j - x_{j-1})$$

Most of the terms in the sum on the right cancel in pairs; that is

$$\sum_{j=1}^n (x_j - x_{j-1}) = (x_1 - x_0) + (x_2 - x_1) + \dots + (x_n - x_{n-1})$$

$$= -x_0 + (x_1 - x_1) + (x_2 - x_2) + \dots + (x_{n-1} - x_{n-1}) + x_n$$

$$= x_n - x_0 = b - a$$

every Riemann sum of f over any partition of

$[a, b]$

Theorem

for f

if f

$[a, b]$

integrable

5.3

Defines

Theorem

if

$[a, b]$

b

$\int_a^b$

a

(49)

$[a, b]$ equals $b-a$ so

$$\int_a^b dx = b-a$$

Theorem Just Statement
for (MCAS)

If f is unbounded on
 $[a, b]$, then f is not
integrable on $[a, b]$.

5.3 Upper and lower Integrals

Defines

Theorem:

If f is bounded on
 $[a, b]$ then

$$\int_a^b f(x) dx \leq \int_a^b f(x) dx$$

Short + Long (50)

Theorem (Imp)

If f is bounded on $[a, b]$ and

$$\int_a^b f(x) dx = \int_a^b f(x) dx = L$$

then f is integrable on $[a, b]$ and

$$\int_a^b f(x) dx = L$$

Proof

If $\epsilon > 0$ there is $\delta > 0$ such that

$$\int_a^b f(x) dx - \epsilon < S(P) \leq S(P) < \int_a^b f(x) dx + \epsilon$$

If $\|P\| < \delta$

If B is a Riemann Sum of f over P then

$$S(P) \leq B \leq S(P)$$

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So

$$|L - \varepsilon| < \delta < \delta < \varepsilon$$

if

$$\|P\| < \delta$$

Theorem (MCAS)

If f is integrable on $[a, b]$ then

$$\int_a^b f(x) dx = \int_a^b f(x) dx = \int_a^b f(x) dx$$

Theorem (MCAS)

If f is bounded on $[a, b]$ then

$$\int_a^b f(x) dx = \int_a^b f(x) dx = L$$

then f is integrable on $[a, b]$

$$\int_a^b f(x) dx = L$$

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Theorem (M(Q))

A bounded function f is integrable on $[a, b]$ if and only if

$$\int_a^b f(x) dx = \int_a^b f(x) dx$$

Theorem (Δ mp)

If f is continuous on $[a, b]$ then f is integrable on $[a, b]$

Theorem (Δ mp)

If f is monotonic on $[a, b]$ then f is integrable on $[a, b]$.

Proof:- from Handouts.

Most Δ mp Theorem

If f is integrable on $[a, b]$ then so is $|f|$ and

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$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

Theorem

If f is integrable on $[a, b]$ and $a \leq a_1 < b_1 \leq b$ then f is integrable on $[a_1, b_1]$

Theorem (Imp)

If f is integrable on $[a, b]$ and $[b, c]$ then f is integrable on $[a, c]$

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

Proof:-

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5.4

Fundamental Theorem of Calculus:-

Lemma:-

If f' is integrable on $[a, b]$ then

$$\int_a^b f'(x) dx = f(b) - f(a)$$

Proof:-

5-4.1

Anti-derivative of a Function

Defines—

A function F is an anti-derivative of f on $[a, b]$ if F is continuous on $[a, b]$ and differentiable on (a, b) with

$$F'(x) = f(x), \quad a < x < b$$

If F is an antiderivative of f on $[a, b]$, then so is $F + c$ for any constant c .

Conversely if F_1 and F_2 are antiderivatives of f on $[a, b]$, then $F_1 - F_2$ is constant on $[a, b]$.

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for every $\varepsilon > 0$ there is
 $\delta_1 > 0$ such that if $|x - x_0| < \delta_1$
is define and

$$|f(x) - f(g(x_0))| < \varepsilon$$

$$\text{if } |x - g(x_0)| < \delta_1$$

Since g is continuous at
 x_0 there is $\delta > 0$ such
that $g(x)$ defined

$$|g(x) - g(x_0)| < \delta_1$$

$$\text{if } |x - x_0| < \delta$$

we can imply that

$$|f(g(x)) - f(g(x_0))| < \varepsilon$$

$$\text{if } |x - x_0| < \delta$$

So

$f \circ g$ is continuous at
 x_0

Example

For the function

$$f(x) = \sqrt{x}$$

$$g(x) = \frac{9-x^2}{x+1}$$

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5.5

Integration by Parts

Theorem:-

u' and v' are integrable on $[a, b]$

$$\int_a^b u(x) v'(x) dx = u(x) v(x) \Big|_a^b - \int_a^b v(x) u'(x) dx$$

$$(uv)' = u'v + uv'$$

integrable " on $[a, b]$

we have

$$\begin{aligned} \int_a^b [u(x) v'(x) + u'(x) v(x)] dx \\ = u(x) v(x) \Big|_a^b \end{aligned}$$

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Theorem

Suppose that f is non-negative and integrable on $[a, b]$ and g is continuous on $[a, b]$

$$\int_a^b f(x)g(x)dx = f(a)\int_a^c g(x)dx + f(b)\int_c^b g(x)dx$$

for some c in $[a, b]$

$$G(x) = \int_a^x g(t)dt$$

then $G'(x) = g(x)$, $a \leq x \leq b$

Therefore by using integration

by Parts with $U = f$ and $V = G$

$$\int_a^b f(x)g(x)dx = f(x)G(x) \Big|_a^b - \int_a^b f'(x)G(x)dx \quad (1)$$

Since f is nonnegative and G is continuous the first mean value theorem of integral implies that

$$(59) \quad \int_a^b f'(x) G(x) dx = G(c) \int_a^b f'(x) dx$$

for some c in $[a, b]$

But we know that

$$\int_a^b f'(x) dx = f(b) - f(a)$$

So,

$$\int_a^b f'(x) G(x) dx = (f(b) - f(a)) \int_a^c g(x) dx$$

Substituting this into eq (1) and noting that $G(c) = 0$

$$\begin{aligned} \int_a^b f(x) g(x) dx &= f(b) \int_a^b g(x) dx - (f(b) - f(a)) \int_a^c g(x) dx \\ &= f(a) \int_a^c g(x) dx + f(b) \left(\int_a^b g(x) dx - \int_a^c g(x) dx \right) \\ &= f(a) \int_a^c g(x) dx + f(b) \int_c^b g(x) dx \end{aligned}$$

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5.6 Integration by Substitution:-

Theorem:-

Suppose the transformation of $x = \phi(t)$ maps interval $c \leq t \leq d$ into the interval $\alpha \leq x \leq \beta$ with $\phi(c) = \alpha$ and $\phi(d) = \beta$ and f is continuous on $[\alpha, \beta]$.

Let ϕ' be integrable on $[c, d]$.

$$\int_{\alpha}^{\beta} f(x) dx = \int_c^d f(\phi(t)) \phi'(t) dt$$

The function $F(x) = \int_{\alpha}^x f(y) dy$ is an antiderivative of f on $[\alpha, \beta]$ and endpoints α and β .

By Fundamental Theorem

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of Calculus

$$\int_{\alpha}^{\beta} f(x) dx = F(\beta) - F(\alpha)$$

By chain rule

$$G(t) = F(\phi(t))$$

is an antiderivative of $f(\phi(t))\phi'(t)$ on $[c, d]$

$$\int_c^d f(\phi(t))\phi'(t) dt$$

$$= G(d) - G(c)$$

$$= F(\phi(d)) - F(\phi(c))$$

$$= F(\beta) - F(\alpha)$$

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Example

Evaluate the integral

$$I = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} (1-2x^2)(1-x^2)^{-1/2} dx$$

Solution:—

$$f(x) = (1-2x^2)(1-x^2)^{-1/2}$$

$$-1/\sqrt{2} \leq x \leq 1/\sqrt{2}$$

$$x = \phi(t) = \sin t \quad -\pi/4 \leq t \leq \pi/4$$

Then

$$\phi'(t) = \cos t$$

$$I = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} f(x) dx = \int_{-\pi/4}^{\pi/4} f(\sin t) \cos t dt$$

$$= \int_{-\pi/4}^{\pi/4} (1-2\sin^2 t)(1-\sin^2 t)^{-1/2} \cos t dt$$

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$$(1 - \sin^2 t)^{-1/2} = (\cos t)^{-1}, \quad -\pi/4 \leq t \leq \pi/4$$

and

$$1 - 2 \sin^2 t = \cos 2t$$

$$I = \int_{-\pi/4}^{\pi/4} \cos 2t \, dt = \left. \frac{\sin 2t}{2} \right|_{-\pi/4}^{\pi/4} = 1$$

Example

Evaluate the integral

$$I = \int_0^{5\pi} \frac{\sin t}{2 + \cos t} \, dt$$